

Fast Fourier Transform:

↳ executes fast DFT
With reduced computation.

$x(n)$... length N sequence.

$$X(k) = \sum_{n=0}^{N-1} x(n) \cdot W_N^{kn}$$

N -pt DFT.

$$0 \leq k \leq N-1$$

$$W_N = e^{-j\frac{2\pi}{N}}$$

$$W_N^{kn} = e^{-j\frac{2\pi}{N}kn}$$

Let $N=4$.

$$X(k) = \sum_{n=0}^3 x(n) \cdot W_N^{kn}, \quad 0 \leq k \leq 3.$$

$$\Rightarrow X(k) = x(0)W_N^{k0} + x(1)W_N^{k1} + x(2)W_N^{k2} + x(3)W_N^{k3}$$

⇒ For each value of k , $\underbrace{\text{no. of complex calculations}}_{\text{multiplications}} = N$..
 $\text{no. of complex additions} = N-1$.

⇒ 1 complex multiplication = 4 real multiplications.

⇒ Now there are 4 values of k , we need

$$\text{In general } \text{no. of complex mult.} = N \times N = N^2$$

$$\text{no. of complex addition} = N(N-1).$$

To minimize these multiplication, we exploit properties of twiddle factor - periodicity & symmetry.

$$\Rightarrow W_N^{k+N} = W_N^k \text{ : periodicity}$$

$$\Rightarrow W_N^{k+\frac{N}{2}} = -W_N^k \text{ : symmetry}$$

$$\Rightarrow W_N^{2k} = e^{-j\frac{2\pi}{N}2k} = e^{-j\frac{2\pi}{N/2}k} = W_{N/2}^k$$

2 algorithms for minimization : DIT FFT, DFF FFT. } use divide & conquer approach.

Here, divide large DFT into half.
Min can be 2-pt DFT.

radix-2 FFT algo ($N = 2^m$)
 $N = \underbrace{2 \times 2 \times 2 \dots \times 2}_{m \text{ times}}$
 $= 2^m$

↑
radix:

Example: $x(n) = \{x(0), x(1)\}$. length: 2.

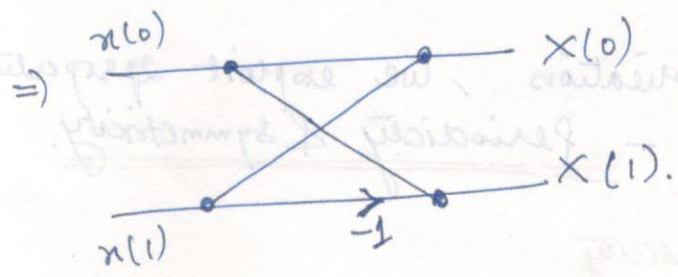
Compute its 2-pt DFT.

$$X(K) = \sum_{n=0}^{N-1} x(n) W_N^{Kn} \quad \underline{0 \leq K \leq 1}$$

$$\Rightarrow X(K) = \sum_{n=0}^1 x(n) W_2^{Kn}$$

$$X(0) = \sum_{n=0}^1 x(n) \cdot W_2^{0n} = x(0) W_2^0 + x(1) W_2^0 = x(0) + x(1)$$

$$X(1) = \sum_{n=0}^1 x(n) W_2^{1n} = x(0) W_2^0 + x(1) \cdot \underbrace{W_2^1}_{(-1)} = x(0) - x(1)$$



: Butterfly diagram
or signal flow graph.

Decimation in Time

DIT - FFT Algorithm ($N=8$). $x(n) \rightarrow N$ -pt sequence. $0 \leq n \leq N-1$, $N=2^m$

$$x(n) = \{x(0) \ x(1) \ x(2) \ x(3) \ x(4) \ x(5) \ x(6) \ x(7)\}$$

DIT
Decimation
in Time

$g(n)$

$$\{x(0) \ x(2) \ x(4) \ x(6)\}$$

$$\{g(0) \ g(1) \ g(2) \ g(3)\}$$

$$0 \leq n \leq 3$$

or

$$0 \leq n \leq \frac{N}{2} - 1$$

$$g(n) = x(2n)$$

$h(n)$

$$\{x(1) \ x(3) \ x(5) \ x(7)\}$$

$$\{\cancel{g(4)} \ \cancel{g(5)} \ \cancel{g(6)} \ \cancel{g(7)}\}$$

$$\{h(0) \ h(1) \ h(2) \ h(3)\}$$

$$h(n) = x(2n+1), \ 0 \leq n \leq 3$$

$$0 \leq n \leq \frac{N}{2} - 1$$

Again decimate in 2-pt DFT

$$\frac{a(n)}{\{g(0) \ g(2)\}}$$

$$= \{x(0) \ x(4)\}$$

$$= \{a(0) \ a(1)\}$$

$$0 \leq n \leq 1$$

$$0 \leq n \leq \frac{N}{4} - 1$$

$b(n)$

$$\{g(1) \ g(3)\}$$

$$= \{x(2) \ x(6)\}$$

$$= \{b(0) \ b(1)\}$$

$$0 \leq n \leq 1$$

$$0 \leq n \leq \frac{N}{4} - 1$$

$c(n)$

$$\{\cancel{g(0)} \ h(2)\}$$

$$= \{x(1) \ x(5)\}$$

$$= \{c(0) \ c(1)\}$$

$d(n)$

$$\{h(1) \ h(3)\}$$

$$\{x(3) \ x(7)\}$$

$$= \{d(0) \ d(1)\}$$

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

$$0 \leq k \leq N-1$$

$$= \sum_{\substack{n=0 \\ n \text{ even}}}^{N-1} x(n) W_N^{kn} + \sum_{\substack{n=0 \\ n \text{ odd}}}^{N-1} x(n) W_N^{kn}$$

Dividing in even & odd indices.

Replace ($n = 2r$)

$$(2r+1), \ 0 \leq r \leq 3$$

$$= \sum_{r=0}^{\frac{N}{2}-1} x(2r) W_N^{kr} + \sum_{r=0}^{\frac{N}{2}-1} x(2r+1) W_N^{k(2r+1)}$$

$$x(2r) = g(r) \quad \& \quad x(2r+1) = h(r)$$

$$\Rightarrow X(k) = \sum_{r=0}^{\frac{N}{2}-1} g(r) W_{N/2}^{kr} + \sum_{r=0}^{\frac{N}{2}-1} h(r) W_{N/2}^{kr} W_N^k$$

$$\Rightarrow \boxed{X(k) = G(k) + W_N^k \cdot H(k)}, \quad 0 \leq k \leq N-1$$

$$(0 \leq k \leq 7)$$

Using symmetric property

$$G(k) = G(k + \frac{N}{2})$$

$$H(k) = H(k + \frac{N}{2})$$

... $\frac{N}{2}$ - point DFTs.

Therefore they will be periodic with the period $N/2$.

* Symmetry $W_N^{k+N/2} = -W_N^k$

$$X(k + \frac{N}{2}) = G(k + \frac{N}{2}) + W_N^{k+N/2} H(k + \frac{N}{2})$$

$$= G(k) - W_N^k H(k)$$

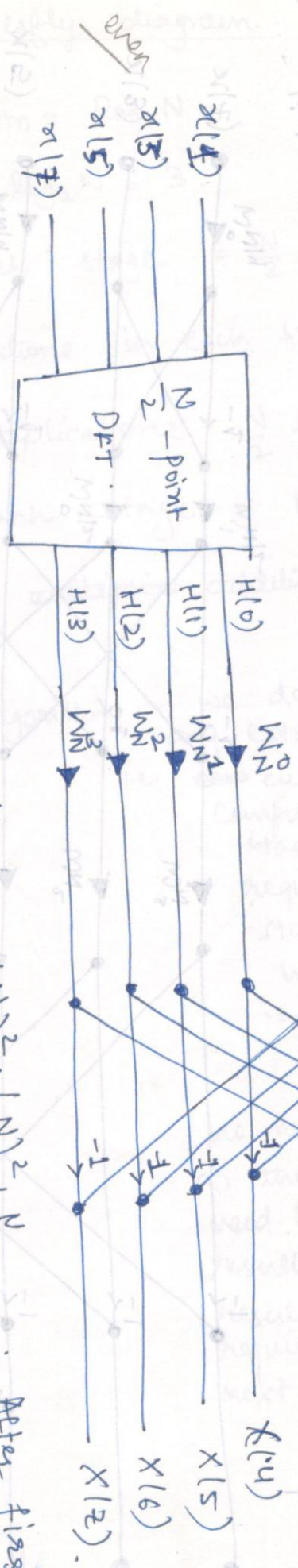
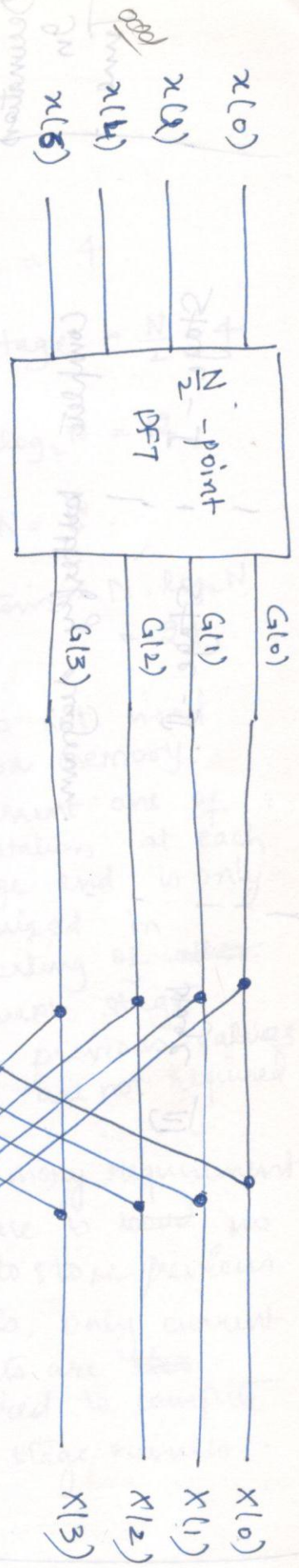
Therefore:

$$X(k) = G(k) + W_N^k H(k), \quad 0 \leq k \leq 3$$

$$X(k + \frac{N}{2}) = G(k) - W_N^k H(k), \quad 0 \leq k \leq 3$$

$$\Rightarrow \begin{cases} X(0) = G(0) + W_N^0 H(0) \\ X(1) = G(1) + W_N^1 H(1) \\ X(2) = G(2) + W_N^2 H(2) \\ X(3) = G(3) + W_N^3 H(3) \end{cases}$$

$$\begin{cases} X(4) = G(0) - W_N^0 H(0) \\ X(5) = G(1) - W_N^1 H(1) \\ X(6) = G(2) - W_N^2 H(2) \\ X(7) = G(3) - W_N^3 H(3) \end{cases}$$



No. of complex multiplications = $\left(\frac{N}{2}\right)^2 + \left(\frac{N}{2}\right)^2 + \frac{N}{2}$

Further process of $\frac{N}{4}$ -pt DFT, to compute $\frac{N}{2}$ -pt DFTs, $G(k)$ & $H(k)$

$a(n) = g(2n)$, $0 \leq n \leq \frac{N}{4} - 1$
 $b(n) = g(2n+1)$, $0 \leq n \leq \frac{N}{4} - 1$

$G(k) = A(k) + W_{N/2}^k B(k)$
 $G(k + \frac{N}{2}) = A(k) - W_{N/2}^k B(k)$
 $0 \leq k \leq \frac{N}{2} - 1$

$a(n) = h(2n)$, $0 \leq n \leq \frac{N}{4} - 1$
 $b(n) = h(2n+1)$, $0 \leq n \leq \frac{N}{4} - 1$

$H(k) = C(k) + W_{N/2}^k D(k)$
 $H(k + \frac{N}{2}) = C(k) - W_{N/2}^k D(k)$
 $0 \leq k \leq \frac{N}{2} - 1$

After first stage decimated

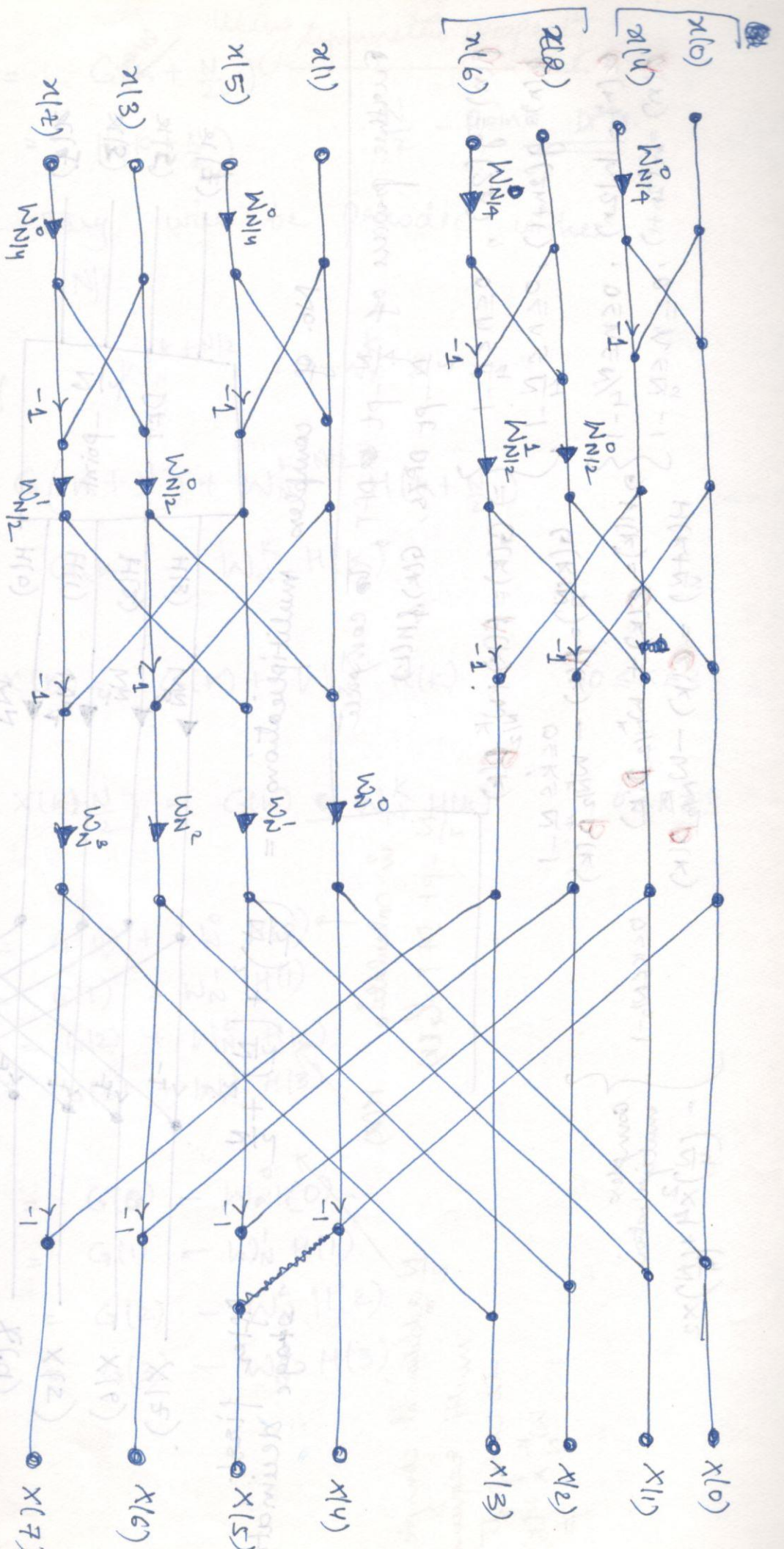
$\frac{N}{2}$ addition of complex

multi requires to compute

$W_N^k \times H(k)$

$N =$

4-Bits



But same or correct order in frequency domain due to DIT

Analysis of complete Butterfly diagram.

$$N = 2^m \Rightarrow m = \log_2 N.$$

1. no. of stages = $m = \log_2 N = 3.$

2. no. of butterflies in each stage = $\frac{N}{2} = 4.$

3. no. of complex multiplications in each stage = $\frac{N}{2} = 4.$

4. Total no. of complex multiplications = $\frac{N}{2} \log_2 N = 12.$

5. no. of ^{complex} additions in each stage = $N = 8.$

6. Total no. of complex ~~multiplications~~ additions = $N \cdot \log_2 N = 24.$

$\Rightarrow$ In place computation/Algorithm - we do not need any extra memory.

$\Rightarrow$ bit-reversal.

i.e. ~~current~~ current ans of computation at each stage end is only required in starting of ~~other~~ next stage.

i.e. previous values are not required.

0	$\rightarrow$	000	$\leftrightarrow$	000	$\rightarrow$	0
1	$\rightarrow$	001	$\leftrightarrow$	100	$\rightarrow$	4
2	$\rightarrow$	010	$\leftrightarrow$	010	$\rightarrow$	2
3	$\rightarrow$	011	$\leftrightarrow$	110	$\rightarrow$	6
4	$\rightarrow$	100	$\leftrightarrow$	001	$\rightarrow$	1
5	$\rightarrow$	101	$\leftrightarrow$	101	$\rightarrow$	5
6	$\rightarrow$	110	$\leftrightarrow$	011	$\rightarrow$	3
7	$\rightarrow$	111	$\leftrightarrow$	111	$\rightarrow$	7

$\downarrow$
Sequence in Time domain.

It is observed that after repeated decimation into even-odd-indexed

subsequences, input sequence gets scrambled. This scrambling is in well-defined order, & can be obtain by bit-reversal method.

no memory requirement as there is ~~need~~ no need to store previous results, only current results are ~~store~~ required to compute next stage results.

DIF - FFT Algorithm

DIT - decimates $x(n)$ into smaller subsequences.

DIF - decimates N -pt DFT sequence $X(k)$ into smaller sequences.

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

divide $x(n)$ into 2 halves

$$= \sum_{n=0}^{N/2-1} x(n) W_N^{kn} + \sum_{n=N/2}^{N-1} x(n) W_N^{kn}$$

$$= \sum_{n=0}^{N/2-1} x(n) W_N^{kn} + \sum_{n=0}^{N/2-1} x(n + \frac{N}{2}) W_N^{k(n + \frac{N}{2})}$$

put $m = n + \frac{N}{2}$ to make summation limits same.

$$= \sum_{n=0}^{N/2-1} x(n) W_N^{kn} + W_N^{kN/2} \sum_{n=0}^{N/2-1} x(n + \frac{N}{2}) W_N^{kn}$$

$$W_N^{kN/2} = e^{j \frac{2\pi}{N} k \cdot \frac{N}{2}} = e^{j\pi k} = (-1)^k$$

$$\Rightarrow X(k) = \sum_{n=0}^{N/2-1} \left[x(n) + (-1)^k x(n + \frac{N}{2}) \right] W_N^{kn}, \quad 0 \leq k \leq N-1.$$

Now decimate $X(k)$ into even & odd-indexed ~~sequences~~ ^{samples}.

$$\Rightarrow X(2k) = \sum_{n=0}^{N/2-1} \left[x(n) + (-1)^{2k} x(n + \frac{N}{2}) \right] W_N^{2kn}, \quad 0 \leq k \leq \frac{N}{2} - 1.$$

and

$$\Rightarrow X(2k+1) = \sum_{n=0}^{N/2-1} \left[x(n) + (-1)^{2k+1} x(n + \frac{N}{2}) \right] W_N^{(2k+1)n}, \quad 0 \leq k \leq \frac{N}{2} - 1.$$

$$\Rightarrow X(2K) = \sum_{n=0}^{N/2-1} \left[x(n) + x(n + \frac{N}{2}) \right] W_{N/2}^{Kn}, \quad 0 \leq K \leq \frac{N}{2} - 1 \quad \text{--- (1)}$$

and

$$X(2K+1) = \sum_{n=0}^{N/2-1} \left\{ \left[x(n) - x(n + \frac{N}{2}) \right] W_N^n \right\} W_{N/2}^{Kn}, \quad 0 \leq K \leq \frac{N}{2} - 1 \quad \text{--- (2)}$$

Let $g(n) = x(n) + x(n + \frac{N}{2}), \quad 0 \leq n \leq \frac{N}{2} - 1$

and $h(n) = \left[x(n) - x(n + \frac{N}{2}) \right] W_N^n, \quad 0 \leq n \leq \frac{N}{2} - 1$

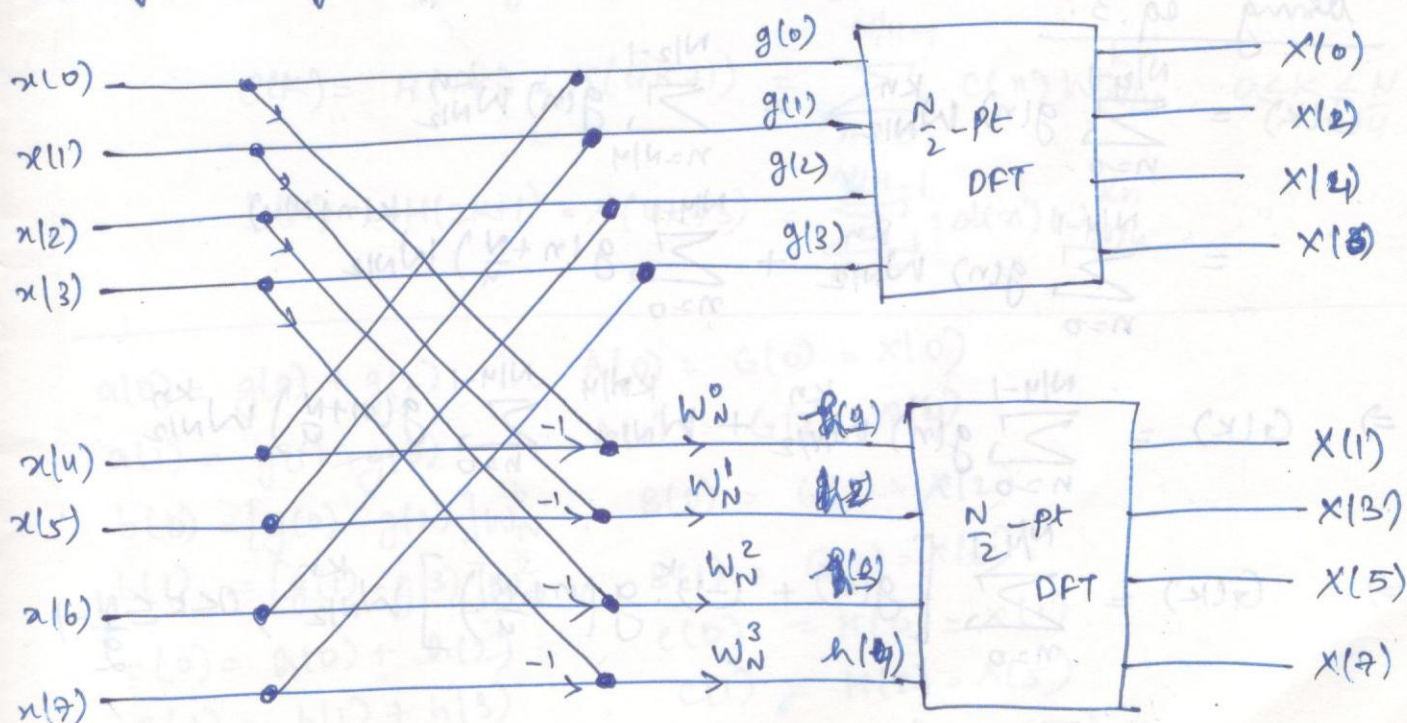
using (1) & (2)

$$\Rightarrow G(K) = X(2K) = \sum_{n=0}^{N/2-1} g(n) \cdot W_{N/2}^{Kn}, \quad 0 \leq K \leq \frac{N}{2} - 1 \quad \text{--- (3)}$$

and $H(K) = X(2K+1) = \sum_{n=0}^{N/2-1} h(n) \cdot W_{N/2}^{Kn}, \quad 0 \leq K \leq \frac{N}{2} - 1 \quad \text{--- (4)}$

Eq. (3) & (4) are $\frac{N}{2}$ -pt DFTs of ~~decomposed~~ $\frac{N}{2}$ -pt DFTs of ~~decomposed~~ N-pt DFT $X(K)$.

Stage I diagram



for $N=8$

$$g(0) = x(0) + x(4)$$

$$G(0) = X(0)$$

$$g(1) = x(1) + x(5)$$

$$G(1) = X(2)$$

$$g(2) = x(2) + x(6)$$

$$G(2) = X(4)$$

$$g(3) = x(3) + x(7)$$

$$G(3) = X(6)$$

$$h(0) = [x(0) - x(4)] W_N^0, \quad H(0) = X(1)$$

$$h(1) = [x(1) - x(5)] W_N^1, \quad H(1) = X(3)$$

$$h(2) = [x(2) - x(6)] W_N^2, \quad H(2) = X(5)$$

$$h(3) = [x(3) - x(7)] W_N^3, \quad H(3) = X(7)$$

Stage - II.

Decimate $\frac{N}{2}$ -pt DFT sequences $G(k)$ & $H(k)$ into $2 \cdot \frac{N}{4}$ pt DFTs each.

using eq. 3.

$$G(k) = \sum_{n=0}^{N/4-1} g(n) W_{N/2}^{kn} + \sum_{n=N/4}^{N/2-1} g(n) W_{N/2}^{kn}$$

$$= \sum_{n=0}^{N/4-1} g(n) W_{N/2}^{kn} + \sum_{n=0}^{N/4-1} g\left(n + \frac{N}{4}\right) W_{N/2}^{k(n+N/4)}$$

$$\Rightarrow G(k) = \sum_{n=0}^{N/4-1} g(n) W_{N/2}^{kn} + W_{N/2}^{kN/4} \sum_{n=0}^{N/4-1} g\left(n + \frac{N}{4}\right) W_{N/2}^{kn}$$

$$\Rightarrow G(k) = \sum_{n=0}^{N/4-1} \left[g(n) + (-1)^k g\left(n + \frac{N}{4}\right) \right] W_{N/2}^{kn}, \quad 0 \leq k \leq \frac{N}{2} - 1$$

$G(k)$ — even & odd indexed sequences.

$$G(2K) = X(4K) = \sum_{n=0}^{N/4-1} \left[g(n) + g\left(n + \frac{N}{4}\right) \right] W_{N/4}^{Kn}, \quad 0 \leq K \leq \frac{N}{4} - 1$$

$$G(2K+1) = X(4K+2) = \sum_{n=0}^{N/4-1} \left\{ \left[g(n) - g\left(n + \frac{N}{4}\right) \right] W_{N/4}^{Kn} \right\} W_{N/4}^{Kn}, \quad 0 \leq K \leq \frac{N}{4} - 1$$

Let $a(n) = g(n) + g\left(n + \frac{N}{4}\right), \quad 0 \leq n \leq \frac{N}{4} - 1$

$b(n) = \left[g(n) - g\left(n + \frac{N}{4}\right) \right] W_{N/4}^{Kn}, \quad "$

$$\Rightarrow A(K) = G(2K) = X(4K) = \sum_{n=0}^{N/4-1} a(n) W_{N/4}^{Kn}, \quad 0 \leq K \leq \frac{N}{4} - 1$$

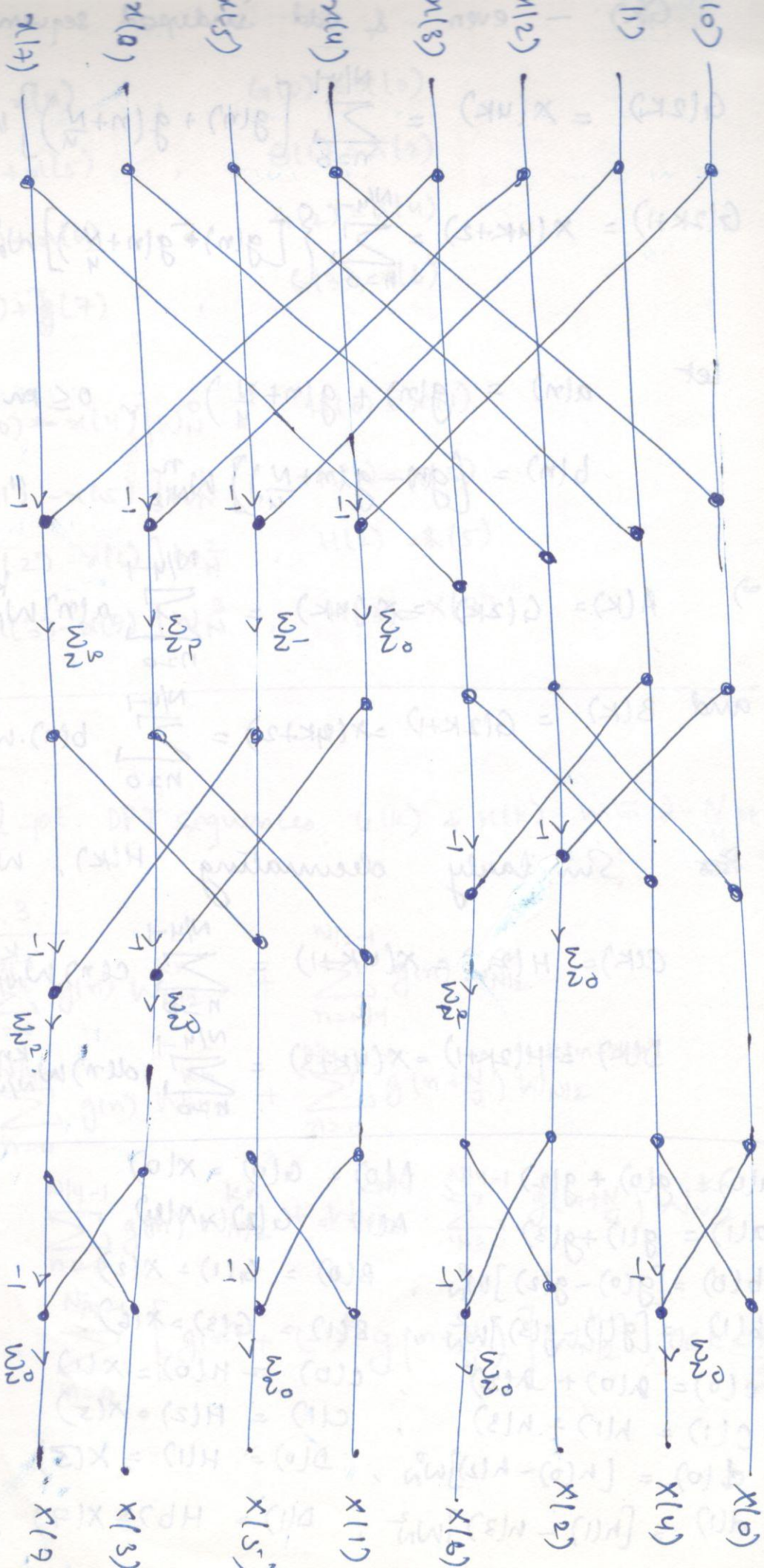
and $B(K) = G(2K+1) = X(4K+2) = \sum_{n=0}^{N/4-1} b(n) W_{N/4}^{Kn}, \quad "$

Similarly decimating $H(k)$, we get.

$$C(K) = H(2K) = X(4K+1) = \sum_{n=0}^{N/4-1} c(n) W_{N/4}^{Kn}, \quad 0 \leq K \leq \frac{N}{4} - 1$$

$$D(K) = H(2K+1) = X(4K+3) = \sum_{n=0}^{N/4-1} d(n) W_{N/4}^{Kn}, \quad "$$

$$\begin{aligned} a(0) &= g(0) + g(2) & A(0) &= G(0) = X(0) \\ a(1) &= g(1) + g(3) & A(1) &= G(2) = X(4) \\ b(0) &= [g(0) - g(2)] W_{N/4}^0 & B(0) &= G(1) = X(2) \\ b(1) &= [g(1) - g(3)] W_{N/4}^1 & B(1) &= G(3) = X(6) \\ c(0) &= a(0) + a(2) & c(0) &= H(0) = X(1) \\ c(1) &= a(1) + a(3) & c(1) &= H(2) = X(5) \\ d(0) &= [a(0) - a(2)] W_{N/4}^0 & D(0) &= H(1) = X(3) \\ d(1) &= [a(1) - a(3)] W_{N/4}^1 & D(1) &= H(3) = X(7) \end{aligned}$$



$g(n)$
 $h(n)$

$a(n)$
 $b(n)$
 $c(n)$
 $d(n)$

$\overline{M}$

Observations from $N=8$ pt DIF-FFT algo Butterfly dia

1. Input is in normal order.
2. o/p is in bit-reversed order.
3. No. of stages $m = \log_2 N = \log_2 8 = 3$.
4. No. of butterflies in each stage $= \frac{N}{2} = 4$.
5. No. of complex ~~mul~~ multiplication in each stage $= \frac{N}{2} = 4$.
6. Total no. of " " $= \frac{N}{2} \cdot \log_2 N = 4 \times 3 = 12$
7. No. of " " additions in each stage $= N = 8$.
8. Total no. of " " $= N \cdot \log_2 N = 8 \times 3 = 24$.